

Cybersecurity Management

University of Minho

Master's Program in Cybersecurity

2025/2026

Practical Lab Exercises #01

I Questions

1. Obtain the ISO/IEC 27000:2018 standard. Focus on Section 3 of the standard. Review the defined concepts and select those you consider most important. For the selected concepts, create a UML Class Diagram. Use various types of relationships—namely association, aggregation, composition, and inheritance—to clearly illustrate how the concepts are related. Pay close attention to multiplicity, and label all the relationships, particularly associations. Do not include attributes or methods unless they pertain to the set of selected concepts. Use an appropriate IT application to design the diagram.
2. Reflect on the concepts defined in the standard. Are there any relevant concepts missing from the standard? If so, explain your rationale and add the missing concepts to your Class Diagram, using red text to distinguish them.

2 Deliverable

The deliverable for this assignment must be submitted as a PDF document formatted for A4 paper size. Each page should be numbered with Arabic numerals, right-aligned in the footer.

The first page of the deliverable must include the following information: name of the course, name of the academic institution, academic year, title of the practical lab exercises, and identification of the students. Student identification should follow the structure: student number, m-dash, full name of the student, m-dash, program the student is enrolled. The identification of each student must appear on a separate line. Students in a group should be listed in alphabetical order by full name of the student.

Each answer must begin on a new page, corresponding to the question number. Answers may be written in Portuguese or English.

Failure to follow these directives will result in a grade of zero for the assignment.

Each student must submit the deliverable individually, using the submission channel provided in the course community on the University of Minho's e-learning platform, by the deadline specified on that platform.

Students who do not submit the deliverable will receive a grade of zero. Submission attempts by other means or after the deadline will be rejected outright.

Prof. Filipe de Sá-Soares
University of Minho, 2025